

mechanism of catalyst

The mechanism of a catalyst operates by offering an alternative reaction pathway with a significantly lower activation energy compared to the uncatalyzed reaction. This reduction in energy barrier accelerates the reaction rate, allowing the process to occur more quickly and often under milder conditions. Catalysts interact with reactant molecules, forming transient intermediates that facilitate the conversion into products. Remarkably, catalysts emerge unchanged after the reaction and can be reused repeatedly. For example, in industrial processes like the Haber process, iron acts as a catalyst to enable nitrogen and hydrogen to combine efficiently into ammonia, a key ingredient in fertilizers. In biological systems, enzymes serve as natural catalysts, driving critical biochemical reactions like digestion and metabolism. These specialized proteins ensure life-sustaining processes occur rapidly and efficiently, highlighting the pivotal role of catalysis in both industrial applications and natural phenomena.